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THE EFFECT OF DIGITAL GAMES AND GAME STRATEGIES ON YOUNG ADOLESCENTS' AGGRESSION

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ABSTRACT

Violence is a major element in many of the present-day's digital games. Despite the extensive research on this subject, the effect of violent digital games on the aggression level among children has not been satisfactorily clarified. The study examines the combined effect of collaborative/competitive game strategy and the presence or absence of violence in the game on the aggression of young adolescents. Fifty-six young adolescents (28 girls, 28 boys) aged 10-13 years old participated in this study. They played digital games and answered questionnaires designed to check the influence of the game on their aggression level immediately after the game. Findings indicate that game strategy may have a significant effect on aggression level: collaborative game strategy was found to repress aggression, whereas competitive strategy was found to raise aggression levels among all subjects. The very existence of violence in the game was not found to be responsible for the aggression level of players. No differences in player aggression levels were found between genders.

INTRODUCTION

Digital games play a key role in today's leisure culture, attracting adults and children alike (Fromme & Unger, 2012; Hamlen, 2011). In the United States, for

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example, it was found that 72% of all households play digital games (Entertainment Association Software, 2011). Various studies in recent years in Western Europe, the United States, Australia, and Israel indicate that over 95% of the youth play digital games on a regular basis (Greenberg, Sherry, Lachlan, Lucas, & Holmstrom, 2010; Henderson, Eshet-Alkalai, & Klemes, 2008; Olson, Kutner, Warner, Almerigi, Baer, Nicholi, et al., 2007). Digital games are popular because they allow virtual experiences in a broad variety of situations and worlds—some imaginary and some representing reality—where players can build new forms of life, try out historical war scenarios, shape characters and locations, play different sports, care for loved animals, build a habitat, and so forth. The current study examined the combined effect of violence in digital games and the use of collaborative or competitive game strategy, on young adolescents' aggression.

In the last decade, alongside the expansion of online communication, we face a change in the perception of the digital game, from one in which a single player competes against the computer, to one in which a dramatically growing number of players participate in social network games, or Mass Multiplayer Online Games (Yee, 2006; Yuan, Hu, Li, & Chu, 2013). For example, in December 2009, some 276 million people regularly played social network games on Facebook (App Leaderboard–Games in: Whon, Lee, Sung, & Bjurnrud, 2010). According to the technology research firm Gartner (2011), the global social game population was 750 million players in 2011 and is expected to double to 1.5 billion by 2015—a compound annual growth rate of 29% (Teufelberger, 2012).

In the research literature, the need for social interaction is reported as one of the major motivations for playing digital games (Cole & Griffiths, 2007): finding friends and joining new friend groups (Griffiths, Davies & Chappell, 2004), creating long-term social contacts between players (Jansz & Tanis, 2007), developing social ties between people of different cultures, bridging social gaps and even replacing face-to-face social encounters (Steinkuehler, 2006, 2007). Other reasons for the attraction to digital games are the love of challenge in a game, an outlet for creativity, fun, thrill, and relief of boredom (Olson, 2010; Olson, Kutner, & Warner, 2008; Olson et al., 2007).

The change in leisure time habits that led to digital games and their central position among youngsters has aroused great interest in recent years among policy makers, educators, and researchers, about the impact they may have on players in particular and on society in general. Since very many of these games contain a prominent element of violence, there is discussion in the field of research regarding the effect such games might have on the asocial aggression of the players (Bushman, Rothstein, & Anderson, 2010; Ferguson & Kilburn, 2010).

In the investigation of the effect of violent digital games on aggression, we can identify two main schools of thought: the first sees the very exposure to violence in digital games as a significant element that over time influences aggression level (Anderson & Bushman, 2002; Bartlett, Harris, & Bruey, 2008); the other emphasizes the role of the environment on shaping aggression, claiming

that to a great extent, aggression is affected by the very fact that the children belong to the "digital games culture"—a society of children who all play violent games as an important part of their lives (Gentile, Lynch, Linder, & Walsh, 2004; Olson et al., 2007).

In this study, we follow Anderson and Bushman (2002), who define aggression as a behavior that is directed toward another person (either virtually or physically) in a purpose of causing harm. In their study of aggression among children, Little, Henrich, Jones, and Hawley (2003) claim that the most common types of aggression among children and adolescents are *Extrovert Aggression*, in which a child attacks another child, either physically or verbally, and *Interpersonal Aggression*, in which a child or a group of children attempt to harm the relationships between other children; for example, by distributing malicious rumors about them in social networks. Bushman and Anderson (2002) oppose this dichotomist distinction of aggression and claim that, in reality, people employ a much more complex range of aggression types. In the present study, the use of the term "aggression level" refers to the level of aggression as measured by different instruments, which are described in the Methods section below.

Following incidents of shooting and murder carried out by adolescents in the United States in the early 2000s, especially the 1999 event in Columbine, in which perpetrators were experts in the violent video game Doom (ID Software, 1993)—and even programmed into it scenes that strongly resembled the actual massacre—research on the effect of violent digital games on aggression in children and adolescents began to flourish. A meta-analysis carried out following these incidents (Anderson & Bushman, 2001) found a significant effect of violent video games on the rise in aggression level. Subsequently, the researchers warned that "exposure to violent video games is a public health threat to children and adolescents" (p. 358). A more moderate approach was presented in the meta-analysis of Sherry (2001) who found little influence of violent digital games on the aggression levels of the players. In reference to the Columbine massacre, Tappan and Kita (1999) had a different viewpoint to that of Anderson and Bushman (2001), noting that human behaviors, aggression among them, are the outcome of a range of personal, social, and cultural elements—including cultural agents such as computer games—that society places at people's disposal. Hence, they claim, examining the effect of violent computer games on aggression in isolation from other cultural-environmental influences does not reflect reality. A similar claim appears in the meta-analysis of Boyle and Hibberd (2005).

Thus, the findings emerging from the research literature regarding the effect of the component of violence in digital games on players' aggression are ambiguous at best. Some studies found that the aggression level rose after the game (Anderson, & Carnagey, 2009; Anderson, Carnaegy, Flanagan, Benjamin, Eubanks, & Valentine, 2004; Anderson & Dill, 2000; Arriaga, Esteves, Carneiro, & Montier, 2006; Gentile et al., 2004; Krcmar & Farrar, 2009). Similar findings are reported in meta-analyses of studies on violent computer games (Anderson,

2004; Anderson & Bushman, 2001). In contrast, other meta-analyses (e.g., Boyle & Hibberd, 2005; Ferguson, 2007; Ferguson & Kilburn, 2009) and many studies (e.g., Bartholow, Sestir, & Davis, 2005; Brady & Matthews, 2006; Elson, 2011; Ferguson, Olson, Kutner, & Warner, 2010; Ferguson & Reuda, 2010; Ivory, & Sriram, 2007; Narvaez & Mattan, 2006; Schmierbach, 2010) were unable to prove any direct impact of violent digital games on aggressive behavior. Ferguson and Kilburn (2010) question the reliability of the meta-analyses examining the effect of violence in digital games on aggression levels, claiming that they are biased because of the selective manner in which the games included were chosen.

Ferguson et al. (2008) and Ivory and Sriram (2007) note that the relationship between aggression and violence in digital games depends on a wide range of factors, the main ones being:

- *Experience and length of play*: Studies examining the effect of the length of time spent playing violent games on aggression (Anderson & Dill, 2000; Bartholow et al., 2005) found greater aggression among experienced, veteran players than among inexperienced ones. A diametrically opposed finding appeared in Ferguson and Reuda (2010), who found a decrease in the aggression level among experienced players compared to novices. A similar finding was obtained among children (Olson et al., 2008) showing that violent digital games calm them down and lower their aggression level.
- *Family background*: On the basis of their findings that a rise in aggression among men who play violent games was found only among players from violent family backgrounds, Brady and Matthews (2006) claim that a violent family background constitutes one of the most significant factors affecting aggression, while the actual violence in games is only a secondary factor.
- *Level of involvement in the game*: Farrar, Krcmar, and Nowak (2006) found a positive correlation between the players' level of involvement in the game and aggression.
- *Level of realism in the game*: Bartlett and Rodeheffer (2009) found that a violent digital game depicting realistic events, that players are familiar with from their real life raised aggression more than a violent digital game depicting imaginary events. In this context, Griffith and Eastin (2009) found that in a digital soccer match showing "real" teams, the defeat of a favorite (virtual) team raised the aggression level among fans more than among those who were not real-life supporters of that team.

The extensive research literature on digital games (e.g., Anderson, Sakamoto, Gentile, Ihori, Shibuya, Yukawa, et al., 2008; Drummond, 2009; Ferguson, 2011; Gentile et al., 2004; Olson et al., 2007, 2008; Zhen, Xie, Zhang, Wang, & Li, 2011) deals comprehensively with the effect of digital games, especially violent ones, on social behavior in general and on the aggression level in particular. However, the information gathered in these studies is based mainly on identifying attitudes through surveys and questionnaires (Olson et al., 2007) and

identifying statistical correlations between digital games and various phenomena such as success at school (Gentile et al., 2004). Surprisingly, in a review of the literature, only one study was found that tested the aggression level among children after they had *actually* played a violent digital game (Polman, De Castro, & Van Aken, 2008) and found that the aggression level after the game rose among the boys but not among the girls. In addition, as far as we know, this is the only study to examine the effect of games on aggression *in the short term* (immediately following the game), while all other studies looked at the *long-term* effect.

Game Strategy and Aggression

The literature on digital games describes a wide range of game-genres, such as racing, sport, action, shooter, role-playing, and simulation (see, for example, Henderson et al., 2008, and the site "All Game Guide"–http://www.allgame.com/). These games can be played in a variety of playing-strategy types (Cole & Griffiths, 2007; Eastin, 2009; Steinkuehler, 2006). The most prominent strategies among adults and adolescents are the *competitive strategy*, in which players compete against each other or against the computer in order to achieve a goal, and the, *collaborative strategy*, in which players collaborate with each other in order to achieve a common goal (Griffiths & Eastin, 2009; Henderson et al., 2008). Because many of the present-days digital games are designed as "dual-strategy games" (i.e., they allow players to play them in both competitive and collaborative strategies), in the current study we chose such "dual-strategy" games in order to investigate the effect of game-strategy on aggression (Griffiths & Eastin, 2009; Henderson et al., 2008).

According to Eastin (2009), in a "dual-strategy game" the choice between a collaborative and a competitive strategy derives from the player's motivation: if it is to achieve a common goal that contributes to everyone in the group, the player will chose to collaborate with the whole group during the game. If it is to acquire all the game's resources at the expense of everyone else, the player will choose a competitive strategy.

As emerges from several studies, choosing a competitive strategy actually lessens the players' chances of attaining their goal since there are many others trying to prevent them from either acquiring resources or winning (Eastin, 2009). According to Berkowitz (1989), when one is prevented from attaining one's goals, the level of frustration rises, and this might lead to a rise in aggression (Anderson & Dill, 2000; Ivory & Sriram, 2007). Hence, many studies claim that the very choice of a competitive game strategy might lead to a rise in the player's aggression (Eastin, 2009). In contrast, with a collaborative game strategy, players help each other to reach a common goal, and the success of one player means success for everyone (Steinkuehler, 2006). Findings from studies that examined the effect of game strategy employed in violent digital games on aggression level (Schmierbach, 2010) are ambiguous. Schmierback (2010) and

Eastin (2009) found that among students, the greater rise in aggression level after playing a violent digital game was found among those who chose a competitive strategy. Other researchers (e.g., Ewoldsen, Eno, Okdie, Velez, Guadagno, & DeCoster, 2012) found that aggression level declined with a collaborative strategy and rose with a competitive strategy.

In the current study, we focused on examining the combined effect of the presence/absence of violence in digital games and of the use of collaborative/competitive game strategy, on the aggression level which was measured immediately after the game, among children between the ages of 10 and 13. We hypothesized that the aggression level among players of a violent game, employing a competitive strategy, would be higher than that of players of a non-violent game, employing a collaborative strategy. Following reports from the research literature, we also assumed that the aggression level would be higher among boys than among girls.

METHODOLOGY

Study Sample

The study examined 56 young Israeli adolescents, 28 boys and 28 girls between the ages of 10 and 13 (average age: 10.58), all from the same upper-middle-class neighborhood and study in the same public school. The participants were recruited as a convenience sample.

Research Tools

The study employed questionnaires to characterize the subjects' game habits and their aggression levels before and after playing.

1. *Questionnaire measuring aggression level* (Anderson, Deuser, & Deneve, 1995): The aim of this questionnaire, which is popular in studies on computer games, is to define the subjects' aggression level at a given point in time through analysis of statements about the person's state of mind. The questionnaire consists of 35 statements describing different states of mind on a scale of 1 (not at all) to 5 (extremely), such as "I feel furious," "I feel I'm about to curse." The questionnaire has a high level of internal consistency: Cronbach's alpha = .96 (Griffiths & Eastin, 2009). It was translated into Hebrew and the translation was validated by three independent language experts. Measurement of the internal reliability of the translated questionnaire showed a high level: Cronbach's alpha = .90.
2. *Game habits questionnaire*: The questionnaire contains background details of the subjects (age, gender, school) and information about their game habits (e.g., experience of playing games on computer and on touch screens, at what age they began to play computer games, what games they like,

and how many hours a week they play). All subjects reported plenty of experience with touch screen games before the start of the study.

3. *Technology platform of the study*: Within the study, the subjects played 3G IPod Touch games (Figure 1). The IPod is a small mobile device with a touch screen and WiFi technology commonly found among children who play Mass Multiplayer Online Games. All subjects in this study had plenty of experience playing games on this device.

The Digital Games

Two simple games with similar graphic displays were chosen, both enabling mastery of their controls after a brief practice and both allowing play with multiple players and on different devices. Neither of the games was known to the players beforehand. Each game enabled players to choose between a competitive and collaborative game strategy.

Violent game: Pocket Legends (Figure 2; 3D MMO; Spacetime, 2011). A multiplayer role play game suitable for children over the age of nine and enabling concurrent participation of thousands of players. In this game, players each choose one of three basic characters (avatars) to represent them: a warrior, an archer, or a potion brewer. Each character has different skills which affect the progress of the game. The environment of the game is a fantasy world where the players fight enemy monsters and win treasures (money, lives, potions). In the competitive game strategy, the winner is the player who kills the most enemies within a 10-minute period. In the collaborative strategy, the players have to



Figure 1. 3G IPod Touch.



Figure 2. Screenshot of the Pocket Legends game.

jointly kill a preset number of enemies within the 10 minutes. Defeat is defined as killing fewer than the target number of the enemy and victory as killing at least that number. Players communicate through mood emoticons (a laughing/frowning etc. smiley), through short set text messages, or by free texting.

Non-violent game: Real *Tennis 2009* (Figure 3; Gameloft S.A., 2010). A multiplayer tennis match enabling simultaneous play on the same court among several players using different devices. The game follows the rules of tennis and is intended for ages of 4 and up (Apple, 2011a, 2011b). Players choose the figure that plays for them out of the avatars of eight real players, and choose the court they will play on. In the competitive game strategy, players must beat each other within a given time. In the collaborative strategy, the players must hit the ball to each other a certain number of times within a given time frame without being "disqualified" (as in the actual rules of tennis). Defeat is defined as not reaching the target number of shots and victory as reaching or passing the preset number of successful shots.

Research Procedure

The study was approved by the university ethics committee and subject participation was approved by the children's parents. The research setup contained four states of play:

- non-violent and collaborative;
- non-violent and competitive;
- violent and collaborative; and
- violent and competitive.



Figure 3. Screenshot of the court in the Real Tennis game.

After providing a general explanation of the research and its procedure and making sure all participants had had experience with the screen and had mastered the game platform, the subjects were all briefly instructed about the games and practiced them for 5 minutes. They were then divided into random pairs and played the two games chosen for the study. Each pair played twice, each time a different game and the opposite game type (for example, if the first game they played was violent and collaborative, the second was non-violent and competitive). There was counterbalancing in the order of the games. Before each game and immediately following it, the subjects completed the aggression level questionnaire. Between games the subjects went out for a 30-minute break—a sufficient length of time for the short-term effect of the digital game to wear off (Bartlett, Branch, Rodeheffer, & Harris, 2009). The games were played so that the players in each pair were in different rooms and played the same game simultaneously either for 10 minutes or until the game ended. The players could speak to each other through the open doorway, but could not see each other except through their virtual characters on the IPod Touch screen. This simulated the natural state of play, where players are not physically in the same location and during the game they observe their avatars and communicate verbally through different means such as Skype, the game software, or telephone.

FINDINGS

The study data were analyzed in two separate moves, in each of which an inter-intra subject variance mixed model analysis was carried out, where the inter-subject variables were violent/non-violent game type and game strategy. The intra-subject variable was aggression level before and after the game. Also

examined were the effect of gender and the correlation between the scope of weekly play and post-play aggression level.

In the examination of the aggression level among the entire sample before and after the game, there was a significant statistical rise in aggression level after playing, compared to the level before playing ($F_{(1,111)} = 4.75$, $p < .05$) with effect size $\eta^2 = .04$. In other words, the very use of the game may have resulted in a rise in the aggression level which was measured when it ended.

As is evident in Table 1, in the analysis of the of the effect of violent/non-violent game type, on the change in aggression level, no statistical difference was found between the appearance of violence in the game and the change in aggression level ($F_{(1,111)} = 2.93$, *NS*; $\eta^2 = .03$). In other words, our findings do not indicate that violence in the game is responsible for a rise in aggression level.

In order to examine the impact of the competitive/collaborative game strategy on the change in aggression level before and after playing a digital game, we conducted an intra-subject analysis (Table 2).

Table 2 shows that game strategy had a medium statistically significant effect on the aggression level ($F_{(1,111)} = 10.93$, $p < .05$, $\eta^2 = .09$). When examining the

Table 1. Aggression Level Before and After Playing Divided by Presence or Absence of Violence

	Aggression level before the game			Aggression level after the game		
Violence in game	*N*	*M*	*SD*	*N*	*M*	*SD*
Non-violent	56	51.45	12.13	56	58.84	24.47
Violent	55	52.62	12.82	55	53.49	16.39
Total	111	52.03	12.45	111	56.19	20.94

Table 2. Changes in Aggression Level by Competitive/Collaborative Game Strategy

	Aggression level before the game			Aggression level after the game		
Game strategy	*N*	*M*	*SD*	*N*	*M*	*SD*
Collaborative	56	52.59	12.71	56	50.82	13.14
Competitive	55	51.45	12.27	55	61.65	25.63
Total	111	52.03	12.45	111	56.19	20.94

interaction between the differences at the effect of the competitive strategy and of the collaborative strategy, it was found that after a competitive game, there was a statistically significant rise in aggression level. In contrast, after a collaborative game, no rise was found in the aggression level. In other words, the collaborative game strategy seems to "suppress" the rise in aggression level that might result from being in a game situation.

We hypothesized that the aggression level of the players who employed a competitive strategy would be higher than for those employing a collaborative strategy. The averages and standard deviations of the changes in aggression level for all four states of play can be seen in Table 3.

As hypothesized, we found a significant difference in the effect on aggression between collaborative and competitive players ($F_{(1,107)} = 9.70$, $p < .05$). The aggression level rose after a competitive game while it did not change after a collaborative game. In addition, we assumed that the effect of game strategy on aggression would be stronger than the effect of the presence of violence in the game. And indeed, in accordance with our hypothesis, no significant difference was found in the change in aggression level between the violent and non-violent games ($F_{(1,107)} = 2.31$, NS). This is in contrast with the effect on aggression found for the competitive/collaborative game strategy. Moreover, no interaction was found between violent/non-violent game and game strategy ($F_{(1,107)} = 0.22$, NS).

To analyze the correlation between intensity of play and aggression level, the research subjects were categorized into "intensive players" (over 11 hours of play a week) and "regular players" (less than 11 hours of play a week) according to the median number of weekly hours of play they reported. No interaction was

Table 3. Changes in Aggression Level for Different States of Play

		Change in aggression level	
Violence in game	Game strategy	Average	*SD*
Violent	Collaborative	–3.57*	10.98
	Competitive	6.20	20.77
	Total (violent)	0.87	16.75
Non-violent	Collaborative	0.31	14.58
	Competitive	13.53*	26.92
	Total (non-violent)	7.39	22.87
Total	Collaborative	–1.77	12.80
	Competitive	10.20	24.38

*Correlation is significant at the 0.05 level (2-tailed).

found between intensity of play and aggression level ($F_{(1,109)} = .78$, *NS*, $\eta^2 = .01$). Similarly, no interaction was found even with the mediation of game type or game strategy. Nor was there any significant difference in aggression level between boys and girls either before or after playing, and there was no effect of the subjects' gender on aggression level ($F_{(1,111)} = .00$, *NS*, $\eta^2 = .00$). In addition, there was no interaction between gender and game type ($F_{(1,111)} = .52$, *NS*, $\eta^2 = .00$) or between gender and game strategy ($F_{(1,111)} = .12$, *NS*, $\eta^2 = .00$).

DISCUSSION

The considerable exposure of youngsters to violent digital games is a cause of great concern to parents, educators, and public figures who fear a rise in aggression resulting from the extensive use of such games. Unfortunately, the recent research literature about the relationships between violent games and aggressive behavior, especially among adolescents, is not conclusive. The current study expands our knowledge about the effect of violent digital games on aggression and proposes a new approach to examine the effect of game strategy and the presence or absence of violence in the game on the aggression level among young adolescents. The finding that the aggression level after playing was higher than before playing for the entire sample, regardless of game type or game strategy, may be explained with the help of models which claim that conflict and competition—prominent features in the game—increase the level of frustration and encourage aggression (Avedon & Sutton-Smith, 1971; Berkowitz, 1989; Zimmerman, 2004). This concurs with the assumptions of the General Aggression Model (Anderson & Bushman, 2002), according to which exposure to a situations encouraging aggression might result in a rise in aggression. Our hypothesis that the presence or absence of violence in the game would affect the degree of change in aggression level after play and that the rise in aggression level would be greater among children who played a violent game than among those who played a non-violent one, was supported by this study. These assumptions were based on the General Aggression Model (Anderson & Bushman, 2002), in which the violent component of the digital games is considered primarily responsible for the rise of aggression, and on studies that found that aggression rose following violent digital games (e.g., Anderson & Dill, 2000; Anderson & Carnagey, 2009; Bartlett & Rodeheffer, 2009).

On the other hand, contrary to our hypothesis, in the current study we found no effect of the presence of violence on the aggression level, something also found in other studies (e.g., Brady & Matthews, 2006; Elson, 2011; Narvaez & Mattan, 2006; Schmierbach, 2010). This finding in our study is of great importance since it emphasizes the limitations of the almost deterministic approach expressed in the General Aggression Model of Anderson and Bushman (2002), which states that the component of violence in a game is the main cause of aggression. Our finding indicates that the change in aggression level following the

use of a violent digital game might be affected by a range of additional factors besides the very violence in the game.

Various researchers (e.g., Bay-Hinitz, Peterson, & Qilitch, 1994) suggest that a collaborative game strategy might reduce aggression level. On the basis of this claim, we hypothesized in this study that the rise in aggression level of players who employed a competitive strategy would be higher than for those employing a collaborative strategy. This hypothesis was supported by our findings, which indicate that after a competitive game there was a significant rise in aggression level, whereas after a collaborative game there was no rise in aggression level. This finding indicates the serious impact the choice of game strategy has on aggression level and the importance of collaborative play in "suppressing" rise in aggression level found when we compared the change in aggression level for the whole research sample without relating to choice of game strategy (Table 3). The findings of our study, regarding the correlation between game strategy and aggression, are supported by the findings in studies conducted on adults (e.g., Eastin, 2009; Ewoldsen et al., 2012; Pipper, O'Brien, Morris, & Winograd, 2006).

The findings of the current study indicate the great importance of a suitable choice of game strategy to regulate aggression levels. On the one hand, the rise in aggression level as a result of any exposure to violent digital games may confirm the fear of educators and parents regarding the negative impact of these games on aggression among children. On the other hand, the significant finding regarding the "calming effect" of collaborative play on the rise in aggression may indicate that with proper guidance, it is possible to obviate the negative effects of violent games—something that requires education and monitoring by parents and educators. This finding illustrates pedagogical implications above and beyond the realm of digital games: it indicates the potential of shared collaborative play activity to regulate aggression even in non-digital media (Deutsch, 1993).

In relating to the implications of this study, one must take into account its following limitations:

1. *Sample size*: The relatively small sample-size ($N = 56$), suggests that future study, on a larger sample, is necessary in order to test the validity of the results.
2. *Limited age-range*: Since this study was conducted on children from a limited age-range (10-13), conclusions should be restricted to this age only. Further study on different age-groups is required in order to test the validity of the results presented here.
3. *Sociodemographic homogeneity*: Participants of this research come from a homogenous upper-middle-class socio-demographic background. Although this homogeneity was an advantage in controlling other mediating variables in the study, further studies, on participants of the same age but from different socio-demographic backgrounds, are necessary.

In summary, future studies, which employ the same research paradigm, but on bigger groups of different ages and sociodemographic backgrounds, are necessary in order to corroborate the findings of this study.

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